Java Complete Guide

Version 1.0 - PrepAI Knowledge Base

1. Introduction

Java is a high-level, object-oriented programming language. It follows the Write Once, Run Anywhere principle using the JVM.

2. Java Architecture

JDK contains development tools, JRE provides the runtime, and JVM executes bytecode.

3. OOP Concepts

Encapsulation, Inheritance, Polymorphism, and Abstraction are the four pillars of OOP.

4. Collections

List, Set, Map, Queue, ArrayList, LinkedList, HashMap, HashSet, TreeMap, PriorityQueue.

5. Exception Handling

Use try, catch, finally, throw and throws to handle exceptions.

6. Multithreading

Thread lifecycle, Runnable, synchronization, ExecutorService.

7. Java 8

Lambda expressions, Stream API, Functional Interfaces, Optional.

8. JDBC

Connect Java applications to relational databases using JDBC drivers.

9. Spring Boot Basics

Auto-configuration, starters, dependency injection, REST APIs.

10. Interview Questions

• What is JVM?
• Difference between JDK and JRE?
• Explain OOP.
• Difference between ArrayList and LinkedList.
• Difference between HashMap and ConcurrentHashMap.
• What is Multithreading?
• What is Exception Handling?
• Explain Stream API.
• What are Lambda Expressions?
• What is JDBC?

11. Coding Tips

Practice clean code, understand time complexity, write unit-testable methods.

12. Practice Problems

Implement CRUD using Java, reverse a string, detect palindrome, sort collections, design a Student Management System.